

SOC ENGINEER PRACTICAL PROJECT

Task 1: Linux and Apache Access Logs to Splunk

Collecting, forwarding and detecting suspicious web traffic in a SIEM

STEP 01

Apache web server running

Apache2 installed on an Ubuntu host and confirmed active and enabled through systemctl.

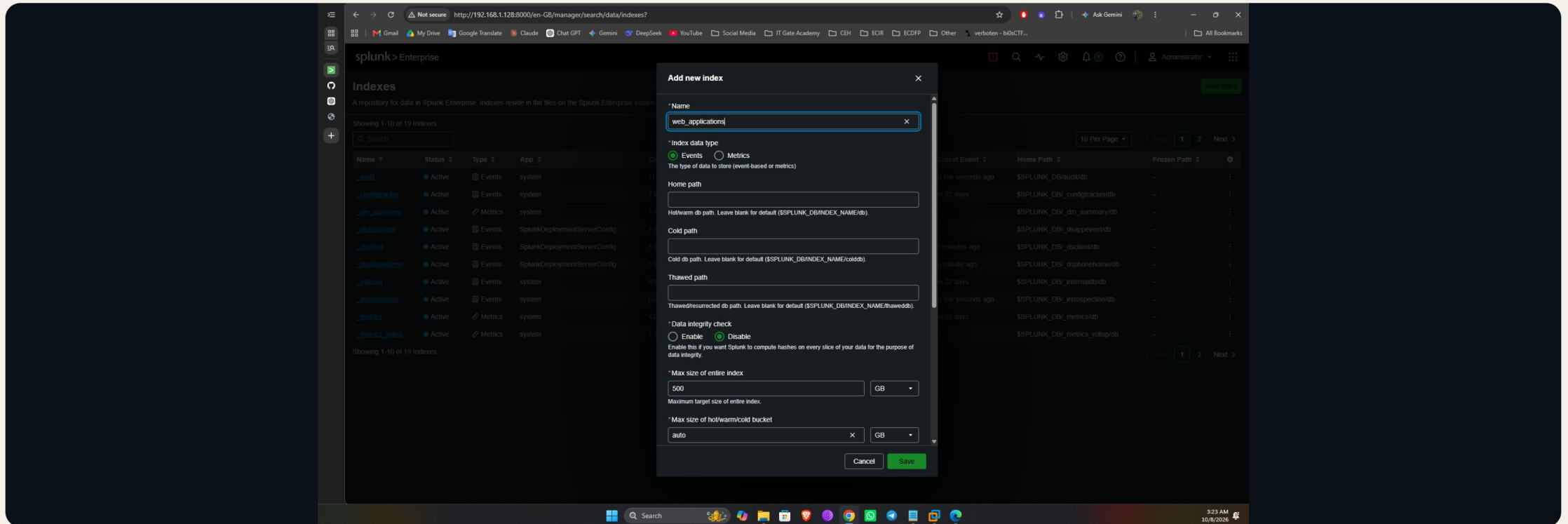
```
root@batman-linux-pc1:/home/batman# sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled)
   Active: active (running) since Thu 2026-10-08 03:07:20 EEST; 52s
     Docs: https://httpd.apache.org/docs/2.4/
    Main PID: 3454 (apache2)
      Tasks: 55 (limit: 3119)
     Memory: 5.2M (peak: 5.7M)
        CPU: 73ms
     CGroup: /system.slice/apache2.service
            └─3454 /usr/sbin/apache2 -k start
              └─3456 /usr/sbin/apache2 -k start
                └─3457 /usr/sbin/apache2 -k start

03:07:20 08 أكت batman-linux-pc1 systemd[1]: Starting apache2.service>
03:07:20 08 أكت batman-linux-pc1 apachectl[3453]: AH00558: apache2: C>
03:07:20 08 أكت batman-linux-pc1 systemd[1]: Started apache2.service >
root@batman-linux-pc1:/home/batman# ip a
```

STEP 02

Dedicated index in Splunk

An events index named web_applications created in Splunk Enterprise to keep web server telemetry separate.



STEP 03

Monitor the Apache access log

Universal Forwarder CLI used to monitor `/var/log/apache2/access.log` and send it to the `web_applications` index.

```
root@batman-linux-pc1:/opt/splunkforwarder/bin# ./splunk add monitor /var/log/apache2
/access.log -index web_applications
Warning: Attempting to revert the SPLUNK_HOME ownership
Warning: Executing "chown -R batman:batman /opt/splunkforwarder"
er- Your session is invalid. Please login.
d Splunk username: admin
.. Password:
Added monitor of '/var/log/apache2/access.log'.
```

STEP 04

Verify inputs.conf

The access log is mapped to web_applications, next to the system logs routed to the linux index.

```
root@batman-linux-pc1:/opt/splunkforwarder/bin# cat /opt/splunkforwarder/etc/apps/search/local/inputs.conf
[monitor:///var/log/syslog]
disabled = false
index = linux

[monitor:///var/log/kern.log]
disabled = false
index = linux

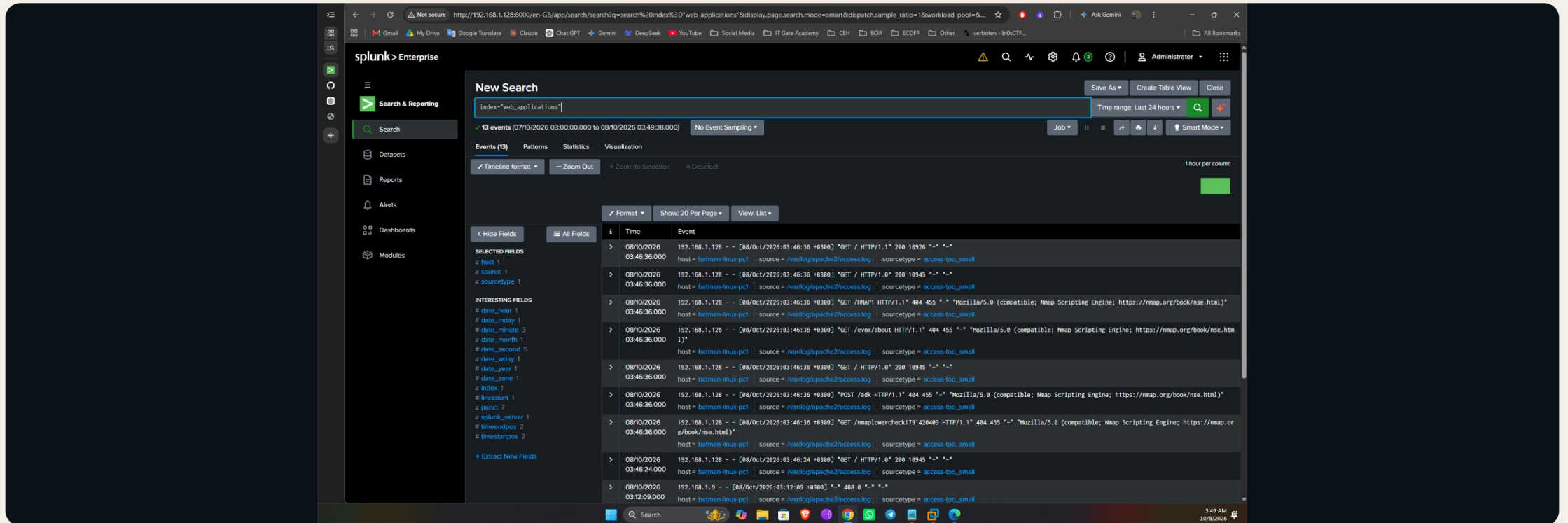
[monitor:///var/log/auth.log]
disabled = false
index = linux

[monitor:///var/log/apache2/access.log]
disabled = false
index = web_applications
```

STEP 05

Scan traffic visible in Splunk

The index holds 13 events. Requests such as /HNAP1 and POST /sdk carry the Nmap Scripting Engine user agent, which exposes the scan.



The screenshot shows the Splunk Enterprise interface. The search bar contains the query `index=web_applications`. The search results are displayed in a table format, showing 13 events. The table has columns for Time, Event, and various fields like host, source, and sourcetype. The events are sorted by time, showing requests from 08/10/2026 03:46:36.000 to 03:12:09.000. The events include various HTTP requests, some of which are identified as Nmap Scripting Engine (NSE) scans.

Time	Event
08/10/2026 03:46:36.000	192.168.1.128 -- [08/Oct/2026:03:46:36 +0300] "GET / HTTP/1.1" 200 10926 "-" "-" host = batman-linux-pc1 source = /var/log/apache2/access.log sourcetype = access-too_small
08/10/2026 03:46:36.000	192.168.1.128 -- [08/Oct/2026:03:46:36 +0300] "GET / HTTP/1.0" 200 10945 "-" "-" host = batman-linux-pc1 source = /var/log/apache2/access.log sourcetype = access-too_small
08/10/2026 03:46:36.000	192.168.1.128 -- [08/Oct/2026:03:46:36 +0300] "GET /HNAP1 HTTP/1.1" 404 455 "-" "-" "Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)" host = batman-linux-pc1 source = /var/log/apache2/access.log sourcetype = access-too_small
08/10/2026 03:46:36.000	192.168.1.128 -- [08/Oct/2026:03:46:36 +0300] "GET /evx/about HTTP/1.1" 404 455 "-" "-" "Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)" host = batman-linux-pc1 source = /var/log/apache2/access.log sourcetype = access-too_small
08/10/2026 03:46:36.000	192.168.1.128 -- [08/Oct/2026:03:46:36 +0300] "GET / HTTP/1.0" 200 10945 "-" "-" host = batman-linux-pc1 source = /var/log/apache2/access.log sourcetype = access-too_small
08/10/2026 03:46:36.000	192.168.1.128 -- [08/Oct/2026:03:46:36 +0300] "POST /sdk HTTP/1.1" 404 455 "-" "-" "Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)" host = batman-linux-pc1 source = /var/log/apache2/access.log sourcetype = access-too_small
08/10/2026 03:46:36.000	192.168.1.128 -- [08/Oct/2026:03:46:36 +0300] "GET /nmaplowercheck1791420403 HTTP/1.1" 404 455 "-" "-" "Mozilla/5.0 (compatible; Nmap Scripting Engine; https://nmap.org/book/nse.html)" host = batman-linux-pc1 source = /var/log/apache2/access.log sourcetype = access-too_small
08/10/2026 03:46:24.000	192.168.1.128 -- [08/Oct/2026:03:46:24 +0300] "GET / HTTP/1.0" 200 10945 "-" "-" host = batman-linux-pc1 source = /var/log/apache2/access.log sourcetype = access-too_small
08/10/2026 03:12:09.000	192.168.1.5 -- [08/Oct/2026:03:12:09 +0300] "-" 408 0 "-" "-" host = batman-linux-pc1 source = /var/log/apache2/access.log sourcetype = access-too_small

STEP 06

Detection alert created

Alert named Nmap Scanning: scheduled hourly, triggers when the number of events is greater than 0.

Alerts

Alerts set a condition that triggers an action, such as sending an email that contains the results of the triggering search to a list of people. Click the name to view the alert. Open the alert in Search to refine the parameters.

This app's ▾

1 Alert

Show: 20 per page ▾

1 ▾

of 1 pages

<

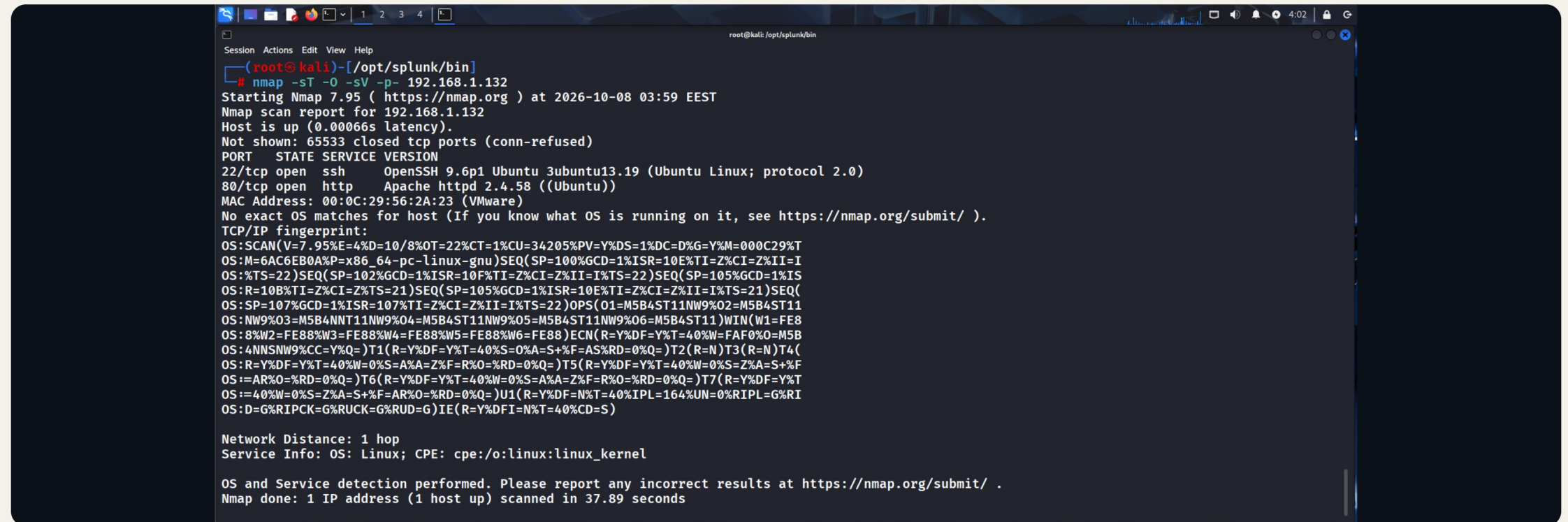
>

Title ↑	Actions	Next scheduled time ↕	Owner ↕	App ↕	Sharing ↕	Status ↕
▼ Nmap Scanning Modified: 8 Oct 2026 03:52:44 Alert type: Scheduled. Hourly, at 15 minutes past the hour. Trigger condition: Number of events is > 0. Actions: ▼ 1 Action Add to Triggered Alerts	Open in search Edit ▾	None	admin	search	Private	✓ Enabled

STEP 07

Full TCP connect scan

A second Nmap scan using TCP connect (-sT) across all 65535 ports from Kali Linux. Ports 22 (SSH) and 80 (Apache 2.4.58) found open.

A screenshot of a terminal window with a dark background. The terminal shows the execution of an Nmap scan from a Kali Linux machine. The command entered is 'nmap -sT -O -sV -p- 192.168.1.132'. The output includes the Nmap version (7.95), the target IP (192.168.1.132), and the scan results. Two ports are identified as open: port 22 (SSH) and port 80 (Apache httpd 2.4.58). The terminal also displays a detailed TCP/IP fingerprint and network distance information. The scan was completed in 37.89 seconds.

```
(root@kali)~/opt/splunk/bin
# nmap -sT -O -sV -p- 192.168.1.132
Starting Nmap 7.95 ( https://nmap.org ) at 2026-10-08 03:59 EEST
Nmap scan report for 192.168.1.132
Host is up (0.00066s latency).
Not shown: 65533 closed tcp ports (conn-refused)
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 9.6p1 Ubuntu 3ubuntu13.19 (Ubuntu Linux; protocol 2.0)
80/tcp    open  http     Apache httpd 2.4.58 ((Ubuntu))
MAC Address: 00:0C:29:56:2A:23 (VMware)
No exact OS matches for host (If you know what OS is running on it, see https://nmap.org/submit/ ).
TCP/IP fingerprint:
OS:SCAN(V=7.95%E=4%D=10/8%OT=22%CT=1%CU=34205%PV=Y%DS=1%DC=D%G=Y%M=000C29%T
OS:M=6AC6EB0A%P=x86_64-pc-linux-gnu)SEQ(SP=100%GCD=1%ISR=10E%TI=Z%CI=Z%II=I
OS:%TS=22)SEQ(SP=102%GCD=1%ISR=10F%TI=Z%CI=Z%II=I%TS=22)SEQ(SP=105%GCD=1%IS
OS:R=10B%TI=Z%CI=Z%TS=21)SEQ(SP=105%GCD=1%ISR=10E%TI=Z%CI=Z%II=I%TS=21)SEQ(
OS:SP=107%GCD=1%ISR=107%TI=Z%CI=Z%II=I%TS=22)OPS(O1=M5B4ST11NW9%O2=M5B4ST11
OS:NW9%O3=M5B4NNT11NW9%O4=M5B4ST11NW9%O5=M5B4ST11NW9%O6=M5B4ST11)WIN(W1=FE8
OS:8%W2=FE88%W3=FE88%W4=FE88%W5=FE88%W6=FE88)ECN(R=Y%DF=Y%T=40%W=FAF0%O=M5B
OS:4NNSNW9%CC=Y%Q=)T1(R=Y%DF=Y%T=40%S=0%A=S+%F=AS%RD=0%Q=)T2(R=N)T3(R=N)T4(
OS:R=Y%DF=Y%T=40%W=0%S=A%A=Z%F=R%O=%RD=0%Q=)T5(R=Y%DF=Y%T=40%W=0%S=Z%A=S+%F
OS:=AR%O=%RD=0%Q=)T6(R=Y%DF=Y%T=40%W=0%S=A%A=Z%F=R%O=%RD=0%Q=)T7(R=Y%DF=Y%T
OS:=40%W=0%S=Z%A=S+%F=AR%O=%RD=0%Q=)U1(R=Y%DF=N%T=40%IPL=164%UN=0%RIPL=G%RI
OS:D=G%RIPCK=G%RUCK=G%RUD=G)IE(R=Y%DFI=N%T=40%CD=S)

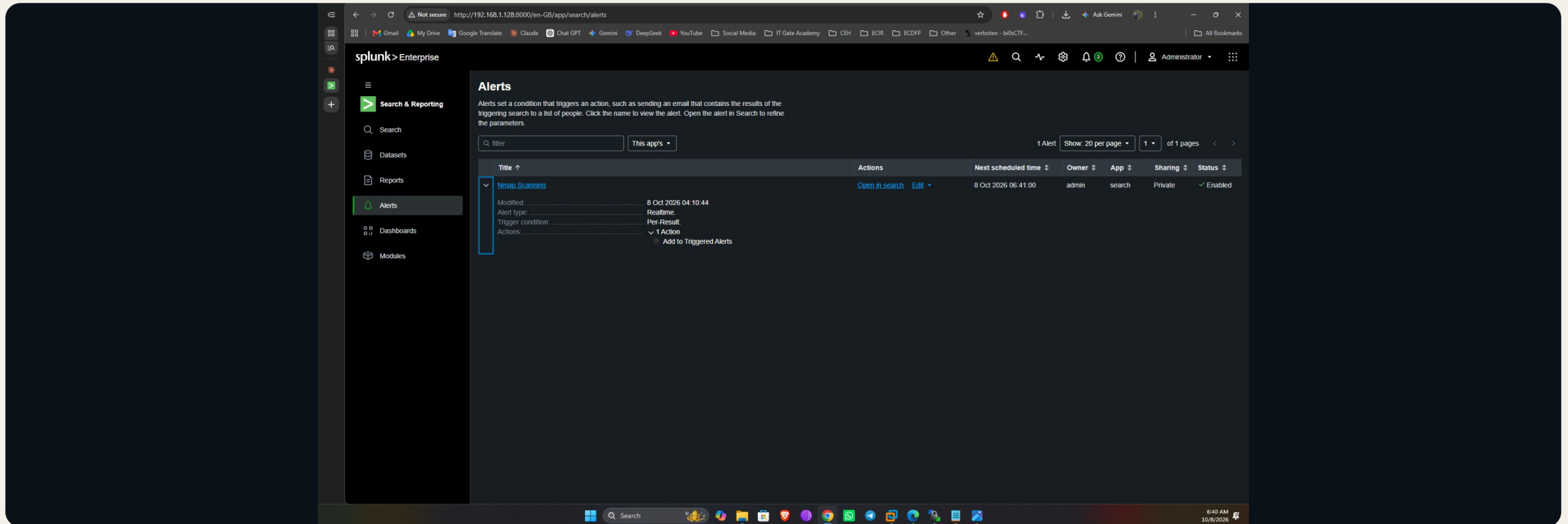
Network Distance: 1 hop
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 37.89 seconds
```

STEP 08

Alert switched to real-time

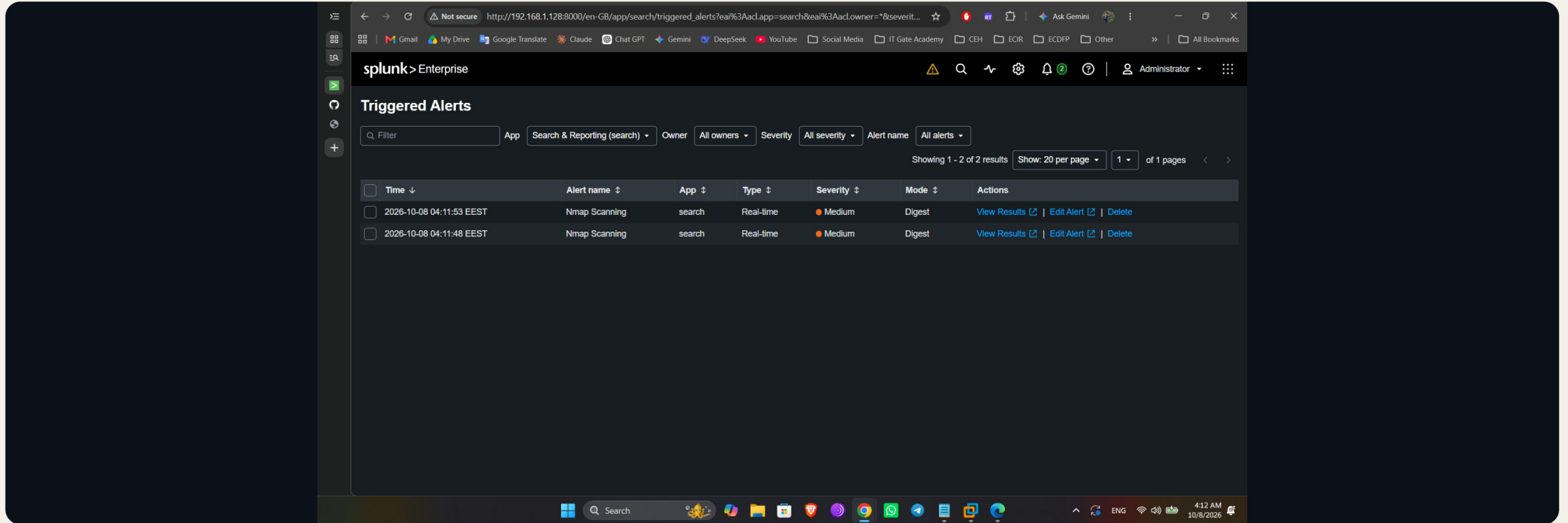
The alert now shows type Realtime with a Per-Result trigger and the Add to Triggered Alerts action.



STEP 09

Alert triggered

Nmap Scanning fired twice within seconds with Medium severity, confirming detection end to end.



The screenshot displays the Splunk Enterprise web interface. The browser address bar shows the URL: `http://192.168.1.128:8000/en-GB/app/search/triggered_alerts?ea%3Aad.app=search&ea%3Aad.owner=*%26severity=...`. The interface title is "splunk>Enterprise". The main heading is "Triggered Alerts". Below the heading, there are filters for "App" (Search & Reporting (search)), "Owner" (All owners), "Severity" (All severity), and "Alert name" (All alerts). The results show 1 - 2 of 2 results, with 20 per page and 1 of 1 pages. The table lists two triggered alerts:

☐	Time ↓	Alert name ↓	App ↓	Type ↓	Severity ↓	Mode ↓	Actions
☐	2026-10-08 04:11:53 EEST	Nmap Scanning	search	Real-time	Medium	Digest	View Results Edit Alert Delete
☐	2026-10-08 04:11:48 EEST	Nmap Scanning	search	Real-time	Medium	Digest	View Results Edit Alert Delete

Result

Collect

Apache access logs captured on Linux, with scan traffic generated.

Forward

Universal Forwarder ships the log to a dedicated Splunk index.

Detect

A real-time alert flags Nmap scanning activity in the web logs.